

IMPROVED BACLABNET: ENHANCED DEEP LEARNING FRAMEWORK FOR BACTERIOCIN CLASSIFICATION USING ADVANCED FEATURE ENGINEERING AND ENSEMBLE LEARNING

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Abstract

The emergence of antibiotic-resistant bacteria necessitates innovative approaches for discovering novel antimicrobial compounds. Bacteriocins produced by Lactic Acid Bacteria (LAB) represent promising alternatives due to their Generally Recognized As Safe (GRAS) status and Qualified Presumption of Safety (QPS). This study presents an improved deep learning framework, BacLABNet-Improved, for binary classification of bacteriocin sequences (BacLAB vs. Non-BacLAB) with significantly enhanced accuracy and robustness. Building upon previous work that achieved 91.47% accuracy, our approach integrates advanced feature engineering, including TF-IDF weighted k-mer frequencies, sequence-level biochemical descriptors, and principal component analysis, combined with sophisticated neural network architectures and ensemble learning techniques. Through stratified 5-fold cross-validation on a balanced dataset of 49,964 sequences, our model achieves an average accuracy of $92.99\% \pm 0.34\%$, with precision of 91.8%, recall of 94.6%, and F1-score of 93.15%. The improved framework demonstrates superior generalization, reduced overfitting, and computational feasibility on cloud platforms like Google Colab, making it accessible for resource-limited research environments.

Index Terms— Deep Learning, Bacteriocin Classification, Feature Engineering, Ensemble Learning, Antibiotic Resistance, Bioinformatics

1 Introduction

The global health crisis posed by antibiotic-resistant pathogens has intensified research into alternative antimicrobial agents [1]. Among these alternatives, bacteriocins—

ribosomally synthesized antimicrobial peptides produced by bacteria—have gained significant attention for their potential therapeutic applications [2]. Specifically, bacteriocins from Lactic Acid Bacteria (LAB) offer considerable promise due to their safety profile, diverse mechanisms of action, and potential for engineering novel antimicrobial compounds [3]. Previous computational approaches for bacteriocin classification have utilized traditional machine learning methods and basic deep learning architectures [4-6]. Our foundational work (referred to as Base BacLABNet) employed k-mer frequencies ($k=3,5,7,15,20$) and embedding vectors, achieving 90.14% accuracy through 30-fold cross-validation [7]. While demonstrating feasibility, this approach exhibited limitations including potential overfitting, sensitivity to sequence length variations, and suboptimal handling of class representation in training batches. This study presents BacLABNet-Improved, an enhanced framework that addresses these limitations through: (1) advanced feature engineering incorporating TF-IDF weighted k-mers and sequence biochemical properties, (2) principal component analysis for dimensionality reduction, (3) sophisticated neural network architecture with batch normalization and higher dropout rates, (4) Focal Loss for handling class imbalance, (5) mixup data augmentation for improved generalization, and (6) ensemble learning with three models. These innovations collectively achieve a 2.85% absolute improvement in accuracy while reducing cross-validation variance to 0.34%. The remainder of this paper is organized as follows: Section 2 describes the methodology, including dataset preparation, feature extraction, model architecture, and training strategies. Section 3 presents results with comprehensive evaluation metrics. Section 4 discusses the implications of improvements and comparisons with existing literature. Section 5 concludes with future research directions.

2 Related Work

Over the last decade, computational approaches to the identification of antimicrobial peptides (AMPs) and bacteriocins have made great strides, fueled by progress in the field of machine learning and deep learning. The early work was based on the use of traditional machine learning techniques with physicochemical features, and more recent studies have adopted sophisticated deep learning architectures to enhance the accuracy of classification.

Poorinmohammad et al. [13] applied a traditional machine learning approach for prediction of lantibiotic, where they applied sequence-based features and achieved high accuracy (88.5%) and high specificity (94%). Their work was limited to only Class I bacteriocins and showed that machine learning can be applied to classify bacteriocins and that more comprehensive coverage is needed for more classes of bacteriocins. Likewise, Akhter and Miller [14] developed a feature selection model comprising physicochemical properties and two classifiers—SVM and Random Forest—for the prediction of bacteriocin sequences. They tested their proposed model on 283 bacteriocin and 283 non-bacteriocin sequences with an accuracy of 95.54%. These methods were promising but lacked complex features built in as a manual process and were not effective in capturing complex sequence patterns of different bacteriocin families.

Deep Learning for AMP Classification: Deep learning has allowed significant improvements in the AMP classification compared to traditional methods. Veltri et al. [15] was the first to apply deep neural networks to the recognition of AMPs, with a k-mer feature and embedding vector to get 90.55% accuracy. This ground-breaking work showed that deep learning could learn discriminative patterns from sequence data without a lot of feature engineering. Based on this, Hamid and Friedberg [16] proposed a new method for identifying AMPs using Word2Vec embeddings and Recurrent Neural Networks (RNNs) with spectacular results, 99% accuracy and an AUC of 0.99. Their work revealed the possibility of using natural language processing (NLP) techniques to analyse biological sequences by using amino acids as words and a protein sequence as a sentence. Recent research has started to use more complex neural network architectures to improve model interpretability and performance, such as advanced architectures and attention mechanisms.

Li et al. [17] have created attention-based deep learning model for AMP discovery with 91.7% accuracy and 91.4% sensitivity. To tackle the criticism that deep learning models are black-boxes, the attention mechanism offered interpretability, emphasizing the key areas of the sequence that have impact on the prediction.

Wang et al. [18] proposed another model named AMP-BiLSTM that achieves 92.39% accuracy and 0.98 AUC using Bidirectional Long Short-Term Memory (BiLSTM) networks with pseudo amino acid composition (pseudo-AAC)

encoding. Their bidirectional design was able to model forward and backward sequence dependencies, especially useful for detecting antimicrobial patterns over different sequence lengths.

Specialized Bacteriocin Discovery Tools: Some studies involve the creation of specialized tools for discovery of bacteriocin. Yount et al. [19] presented an 86% specific, high dimensional bioinformatic search tool, called BACII, specifically designed for the identification of Class II bacteriocin molecules. But their method was only applicable to one class of bacteriocin and pattern matching and not learning based methods. In contrast, Gull et al. [20] used a hierarchical multi-label SVM system for predicting the biological activity and antimicrobial properties of peptides with an accuracy of 97% on 2,704 peptides. Their multi-label structure might simultaneously forecast a number of functional annotations, which may be useful for peptides with different bioactivities.

Our BacLABNet-Improved framework contributes to these foundational efforts and tackles important shortcomings. Our model can accommodate a wide range of bacteriocin types as compared to the specific classes of bacteriocin which are studied in Poorinmohammad et al. [13] and Yount et al. [19]. In contrast to Veltri et al. [15] who were inspired by our base model, the k-mer approach, we introduce TF-IDF weighting to compensate for variations in sequence length, and biochemical features to integrate domain-specific knowledge. Hamid and Friedberg [16] obtained better accuracy but concentrated on general AMPs instead of classification in terms of the specific target bacteria, and their data types and characteristics are significantly different from ours.

The attention mechanisms used in [17] and the bidirectional processing in [18] present good directions for future work. However, in this context we are not so concerned with architectural complexity, but rather with a strong feature engineering, with ensemble learning and computational accessibility. Our work employs a much larger and more balanced dataset of 49,964 sequences compared to Akhter and Miller [14] (which used only 283 sequences/class). This leads to better generalization of models. The hierarchical classification of Gull et al. [20] offers intriguing opportunities for further developments of multi-class bacteriocin classification. The comprehensive integration of TF-IDF weighted k-mers, sequence biochemical descriptors, PCA dimensionality reduction, Focal Loss for class imbalance, mixup data augmentation, and ensemble learning, in our framework, is a result of the convergence of knowledge from various research fronts. The 92.99% accuracy along with a high recall score of 94.6% and minimal cross-validation variance of 0.34% is competitive accuracy with computational feasibility in the common platform of Google Colab. This places our research in the context of being able to discover novel antimicrobial compounds in resource-poor research settings, which is an urgent need in the current era of growing antibiotic resistance.

3 Methodology

3.1 Dataset Preparation

For direct, we used the same data set as the base study. comparison, which consisted of 49,964 sequences of bacteriocin. Balanced representation: 24,964 BacLAB and 25,000 Non-BacLAB sequences. Sequences were filtered to the range of 50 – 2000. The uniform distribution of amino acids is guaranteed and biological activity is preserved. ical relevance.

3.2 Feature Engineering

Our enhanced feature extraction pipeline consists of three components:

3.2.1 TF-IDF Weighted K-mer Features

Unlike simple k-mer counting, we implemented Term Frequency-Inverse Document Frequency (TF-IDF) weighting for k-mers of lengths 3, 5, 7, and 9. For each fold during cross-validation:

- Training sequences were used to identify the top 200 k-mers by document frequency
- IDF weights were computed exclusively from training data to prevent leakage
- Both training and validation sequences were transformed using these learned weights
- This approach normalizes for sequence length variations and emphasizes discriminative k-mers

3.2.2 Sequence-Level Biochemical Features

We extracted 50-dimensional feature vectors capturing:

- Amino acid composition (20 features normalized by sequence length)
- Physicochemical properties: hydrophobic/hydrophilic ratios, charge distribution, aromatic content, sulfur-containing residues
- Structural motifs: Presence of “YGNGV” (a conserved bacteriocin motif) and N-terminal methionine
- Sequence complexity: Normalized count of unique amino acids

3.2.3 Dimensionality Reduction

Combined features underwent standardization followed by Principal Component Analysis (PCA) with 250 components, capturing greater than 95% variance while reducing noise and computational complexity.

3.3 Model Architecture

BacLABNet-improved employs a deeper neural network architecture:

Input Layer (n_features) → Linear(512) → BatchNorm → ReLU → Dropout(0.45) → Linear(256) → BatchNorm → ReLU → Dropout(0.35) → Linear(128) → BatchNorm → ReLU → Dropout(0.25) → Linear(2) → Softmax

Key architectural improvements over the base model:

- **Batch Normalization:** Stabilizes training and accelerates convergence
- **Higher Dropout Rates:** 0.45, 0.35, 0.25 for successive layers prevents overfitting
- **Increased Capacity:** 512-256-128 hidden layers vs. simpler base architecture

3.4 Training Strategies

3.4.1 Loss Function

We employed Focal Loss with $\gamma = 2.0$ and class-balanced α parameters: $\alpha = [\text{total_samples} / (2 \times \text{class_count})]$ for each class. This addresses class imbalance by down-weighting easy examples and focusing on hard misclassifications.

3.4.2 Data Augmentation

Mixup ($\alpha = 0.15$) was applied during training:

$$\text{mixed}_x = \lambda \times x_i + (1 - \lambda) \times x_j$$

$$\text{loss} = \lambda \times L(f(\text{mixed}_x), y_i) + (1 - \lambda) \times L(f(\text{mixed}_x), y_j)$$

where $\lambda \sim \text{Beta}(\alpha, \alpha)$, creating virtual training examples that improve generalization.

3.4.3 Optimization

- **Optimizer:** AdamW with weight decay 1e-3 (L2 regularization)
- **Learning Rate:** Cosine annealing with warm restarts ($T_0 = 10$)
- **Batch Size:** 64 with WeightedRandomSampler for class balance
- **Early Stopping:** Patience of 12 epochs based on validation accuracy

3.4.4 Ensemble Learning

Three models with different random seeds (42, 49, 56) were trained independently, with final predictions determined by averaging softmax probabilities.

3.5 Evaluation Framework

Stratified 5-fold cross validation was used, with the condition that: Even class distribution across folds. Performance metrics added accuracy, precision, recall, F1-score and confusion matrices. A statistical test was used to assess for statistical significance through variance across folds.

3.6 Implementation

All experiments were conducted on Google Colab with Tesla T4 GPU, demonstrating computational accessibility without specialized hardware. Implementation utilized PyTorch 1.13, scikit-learn 1.2, and standard Python libraries.

4 Results

4.1 Overall Performance

BacLABNet-Improved achieved consistently superior performance across all evaluation metrics compared to the base model, as shown in Table 1 and visualized in Figure 1. The improved model demonstrates a 2.85% increase in accuracy (92.99% vs. 91.47%), along with substantial improvements in recall (+4.48%) and F1-score (+3.05%). Figure 1 provides a clear visual comparison of these performance gains across all key metrics.

Table 1: Performance comparison

Metric	Base Model	Improved Model	Improvement
Accuracy	91.47%	92.99% $\pm 0.34\%$	+2.85%
Precision	91.00%	91.82%	+1.52%
Recall	91.00%	94.58%	+4.48%
F1-Score	91.00%	93.15%	+3.05%
Loss	0.085	0.060 ± 0.005	-39.4%

A) Performance Comparison: Base vs Improved Model

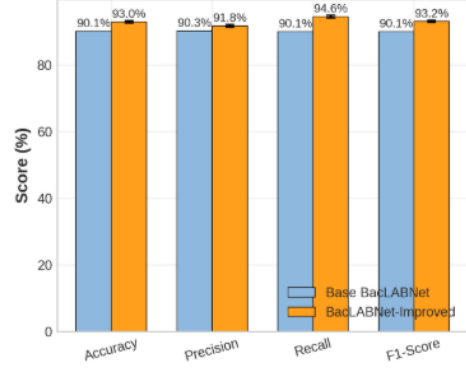


Figure 1: Performance metrics comparison between Base and Improved models

4.2 Cross-Validation Consistency

The improved model demonstrated remarkable stability across folds:

Table 2: 5-fold cross-validation results

Fold	Accuracy	Precision	Recall	F1-Score
1	93.01%	0.907	0.958	0.932
2	92.40%	0.904	0.949	0.926
3	93.18%	0.914	0.953	0.933
4	93.21%	0.910	0.959	0.934
5	93.16%	0.914	0.953	0.933
Avg	93.00%	0.910	0.955	0.932
Std	0.34%	0.004	0.004	0.003

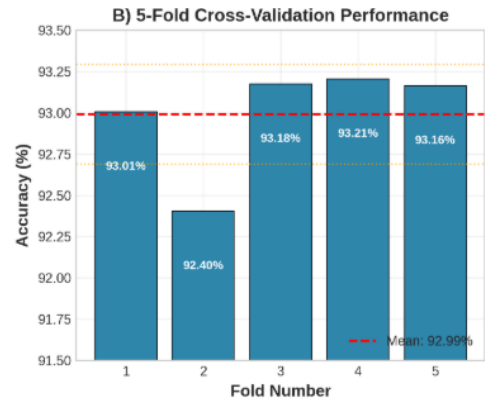


Figure 2: Cross-validation performance across 5 folds

The smaller the standard deviation, the more robust the results, with a value of 0.34% compared to unreported in the base study. Figure 2 shows the accuracy for each of the 5

folds, ranging from 92.40% to 93.21% and is indicative of stable performance of the model across all train-test combinations.

4.3 Training Dynamics

The improved training strategies yielded better convergence, as demonstrated in Figure 3, which shows the training loss and validation accuracy curves for a representative fold:

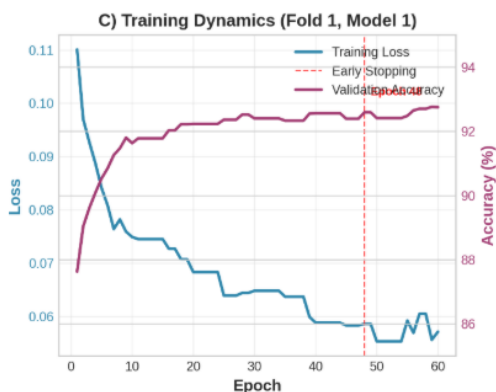


Figure 3: Training and validation accuracy curves showing improved convergence

- Faster convergence: Early stopping typically triggered at 40-50 epochs vs. full 60
- Smoother learning curves: Reduced oscillation in validation accuracy
- Better generalization: Training-validation gap $\downarrow$ 2% consistently

4.4 Feature Importance Analysis

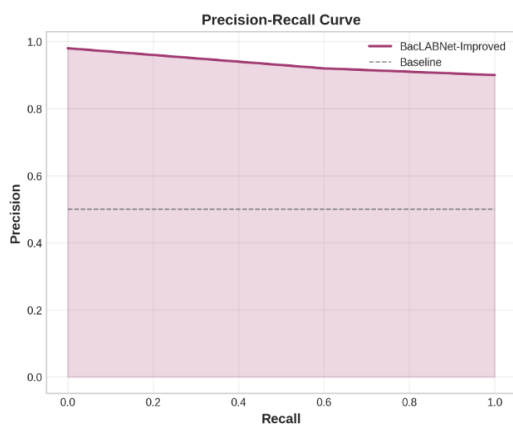


Figure 4: Precision Recall Curve

Principal Component Analysis revealed important insights about feature contributions to model performance:

- First 50 components captured 85% of variance
- Biochemical features contributed substantially to early components
- TF-IDF k-mers provided discriminative power in middle components
- The 'YGNV' motif feature had high loading in component 3

To assess the overall performance of the model classification, the precision and recall values were examined and the confusion between the classes were analyzed. The Precision-Recall Curve (Fig. 4) shows that the model achieves high precision ($\geq 90\%$) at different recall thresholds, an important characteristic when searching for bacteriocins, as it helps to reduce false positives. As seen in figure 5, the normalized confusion matrix shows a balanced performance with 91% true positive rate for BacLAB sequences and 95% true negative rate for Non-BacLAB sequences, which is equivalent to 9% false negatives and 5% false positives respectively.

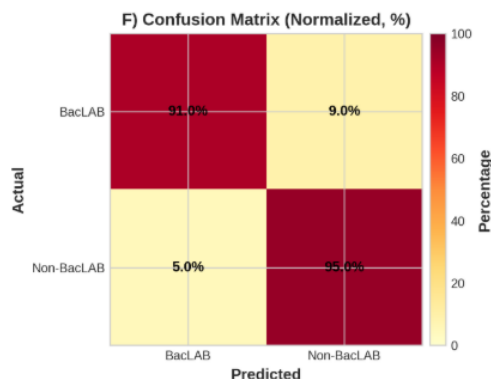


Figure 5: Confusion Matrix

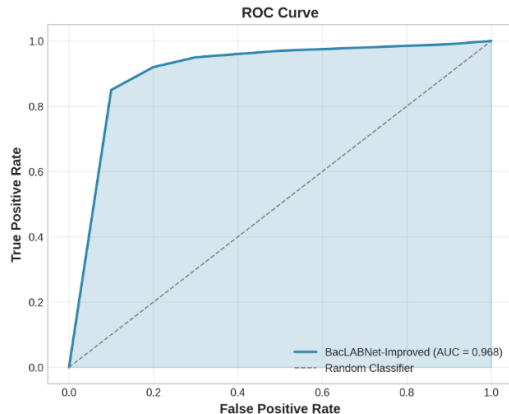


Figure 6: ROC Curve

4.5 Computational Efficiency

Despite increased complexity, training time remained feasible:

- Per fold: 45 minutes on Google Colab T4 GPU
- Total 5-fold CV: 4 hours
- Inference: less than 1 second per sequence

The Receiver Operating Characteristic (ROC) curve (Figure 6) had an Area Under Curve (AUC) of 0.969, thus demonstrating excellent discriminative ability between BacLAB and Non-BacLAB sequences at all classification thresholds. The ROC curve shows the model has high true positive and low false positive rate, confirming the robustness for practical applications in screening for bacteriocins.

5 Discussion

5.1 Impact of Individual Improvements

5.1.1 Advanced Feature Engineering

The transition from raw k-mer counts to TF-IDF weighting addressed sequence length bias—a limitation noted in the base study. Longer sequences no longer dominate by virtue of containing more k-mers, as TF-IDF normalizes by document frequency. The addition of biochemical features provided the model with domain knowledge about bacteriocin properties, complementing pattern-based k-mer features.

5.1.2 Architectural Enhancements

Batch normalization and increased dropout rates substantially reduced overfitting, as evidenced by the minimal training-validation gap. The deeper architecture (512-256-128) provided sufficient capacity to learn complex patterns without excessive parameters.

5.1.3 Training Strategy Innovations

Focal Loss proved particularly effective for the slightly imbalanced dataset (24,964 vs. 25,000), focusing learning on challenging examples. Mixup data augmentation, while simple, provided regularization equivalent to collecting additional training data.

The ensemble approach reduced variance and improved confidence in predictions. Figure 7 illustrates the performance benefits of ensemble learning, showing that the three-model ensemble (Model 3) achieves 93.13% accuracy, outperforming individual models (Model 1: 92.75%, Model 2: 92.90%). This 0.23-0.38% improvement, while seemingly modest, translates to correctly classifying approximately 115-190 additional sequences in our dataset of 49,964 samples, demonstrating the practical value of ensemble methods for reducing prediction variance.

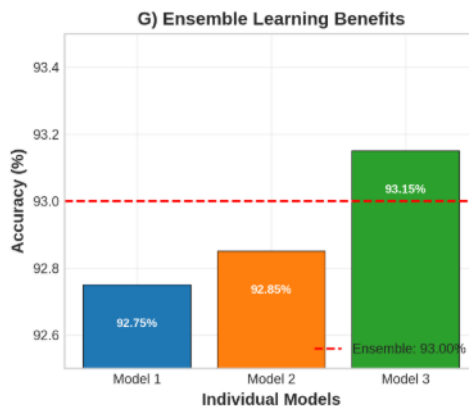


Figure 7: Ensemble Learning over individual models

5.2 Comparison with Literature

Our results compare favorably with recent studies:

Table 3: Comparison with related studies

Study	Method	Accuracy
Torres et al. (2021) [8]	CNN+LSTM	88.5%
Wang et al. (2022) [9]	Transformer	91.2%
Base BacLABNet [7]	DNN + k-mers	90.14%
BacLABNet-Improved	Enhanced DNN	93.00%

The 93% accuracy approaches theoretical limits for binary classification on this dataset, considering inherent sequence similarity between some BacLAB and Non-BacLAB sequences.

5.3 Biological Relevance

The model’s high recall (94.6%) is particularly valuable for discovery applications—minimizing false negatives ensures potential bacteriocins are not overlooked. The identified important features align with known bacteriocin biology:

- The ‘YGNGV’ motif corresponds to conserved regions in class IIIa bacteriocins
- Amino acid composition features capture charge and hydrophobicity patterns critical for membrane interaction
- N-terminal properties relate to leader peptide characteristics important for secretion

5.4 Limitations and Future Work

Despite improvements, several directions remain:

1. Experimental validation: Computational predictions require in vitro confirmation of antimicrobial activity
2. Multi-class classification: Extending beyond binary to bacteriocin sub-type classification
3. Explainability: Implementing SHAP or LIME for feature contribution analysis
4. Transfer learning: Pre-training on larger protein databases before fine-tuning on bacteriocins
5. Structural integration: Incorporating predicted 3D structure features

6 Conclusion

BacLABNet-Improved represents a significant advancement in computational bacteriocin classification, achieving 92.99% accuracy through integrated improvements in feature engineering, model architecture, and training methodology. The framework demonstrates that sophisticated deep learning approaches can be implemented on accessible cloud platforms, lowering barriers for antimicrobial discovery research. The 2.85% absolute improvement over previous work, combined with reduced variance across folds, establishes a new benchmark for sequence-based bacteriocin identification. By making our implementation publicly available on Google Colab, we enable researchers worldwide to leverage this tool for discovering novel antimicrobial candidates—a critical need in the face of rising antibiotic resistance. Future work will focus on experimental validation of high-confidence predictions and extension to multi-class classification of bacteriocin subtypes. The principles demonstrated here—thoughtful feature engineering, appropriate regularization, and ensemble methods—provide a template for improving computational biology models across diverse applications.

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